Haydeé Contreras Peruyero

Postdoctoral Research at Centro de Ciencias Matemáticas UNAM

18 Aug 1989

CCM UNAM

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https://haydeeperuyero.github.io/

haydeeperuyero@matmor.unam.mx

About me

My research interests lie at the intersection of geometry, topology, and bioinformatics. My work has focused on geometric group theory and the application of Topological Data Analysis (TDA) to problems in genomics, metagenomics, and pangenomics. I also use machine learning and statistics to explore structures in biological data. I have a strong passion for teaching, programming, and learning languages.

Skills

- Math thinking
- Team worker
- Bioinformatics coding skills
- SQL, PowerBI, Conda
- R, Python, Git&Github, Bash
- Teaching
- Written English
- Oral English
- Project managment

(*)[The skill scale is from 0 (Fundamental Awareness) to 6 (Expert).]

Interests

Topological ang Geometric Data Analysis | Bioinformatics | Geometric GroupTheory | Global Riemannian Geometry | Quantum Geometry and Topology

Education

2016-2021 Ph. D. in Science in Mathematics — Instituto de Matemáticas, UNAM
Thesis:*Quasi-Isometric Rigidity in Higher Graph Manifolds*

2020-2021 Specialty in Applied Statistics — Instituto de Investigaciones en Matemáticas Aplicadas y en Sistemas

2014-2016 M.Sc. in Mathematics — Instituto de Matemáticas, UNAM
Thesis: On the Geometry and Topology of Graph Manifolds

2007-2012 B.Sc. in Mathematics — Universidad Veracruzana
Thesis: Triangulations of Hyperbolic 3-Manifolds Fibered over the Circle with Torus with a Hole as Fiber

Professional Experience

2025- Postdoctoral Research — CONAHCYTH, CCM UNAM
Topological Data Analysis Applied to Genomic Data and Cryptography with Geometric Group Theory

2023-2025 Postdoctoral research DGAPA — CCM UNAM
Applications of Topological Data Analysis and Quasi-Isometry Invariants

2022-2023 Postdoctoral research CONACyT Ciencia de Frontera — CONACyT, CCM UNAM
Invariant Properties of Higher Graph Manifolds

2021-2022 Head of the Department of Basic Sciences — ITSMT

Relevant Publications for the Position

2025 Topological data analysis captures Horizontal Gene Transfer in Antimicrobial Resistance gene families among clinically relevant bacteria. Guerrero-FLores et al. Front. Microbiol. 16:1461293. https://doi.org/10.3389/fmicb.2025.1461293

2024 Multidisciplinary Workshop on Pangenomics and Topological Data Analysis, H. Contreras-Peruyero et al. Bioinformatics Advances, 2024;, vbae139 https://doi.org/10.1093/bioadv/vbae139

2024 CAMDA 2023: finding patterns in urban microbiomes. H. Contreras-Peruyero et al. Front. Genet. (2024) 15:1449461. https://doi.org/10.3389/fgene.2024.1449461

2024 Physicochemical Properties of Two Mexican Stingless Bee Honeys to Strengthen Their Biocultural Value. A. Xolalpa-Aroche et al. (2024). Revista Brasileira de Farmacognosia, 1-11. https://doi.org/10.1007/s43450-024-00566-z

2024 Actinobacteria bioprospecting from ore-forming environments. C. Aguilar et al. Microbial Genomics, 10(5), 001253. ISNN 2057-5858, https://doi.org/10.1099/mgen.0.001253

Other Publications

2025 Asymptotic dimension and geometric decompositions in dimensions 3 and 4, H. Conteras Peruyero and P. Suárez-Serrato. Journal of the Australian Mathematical Society. Published online 2025:1-26. https://doi.org/10.1017/S1446788725000072

2025 Linear bounds for the lengths of geodesics on manifolds with curvature bounded below. Accepted April 2025. https://arxiv.org/abs/2410.10975

2022 Collapsing and group growth as obstructions to Einstein metrics on some smooth 4-manifolds. H. Contreras Peruyero and P. Suárez-Serrato, New York Journal of Mathematics 28 (2022), 659–671. http://nyjm.albany.edu/j/2022/28-27.html

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Accepted papers

2025 Fractal dimensions of complex networks: advocating for a topological approach. Andreeva, R. et al. To apper at the AWM-IMA Springer volume "Research in Computational Topology". https://arxiv.org/abs/2506.15236. Accepted May 2025.

Preprints

2025 New Acetilactobacillus species is the most abundant microorganism in the microbiome of native stingless bee honeys in Mexico. Submitted June 2025.

2024 Lengths of the Orthogonal Geodesic Chords on Riemannian Manifolds. In preparation.

2022 On the quasi-isometric rigidity of chambers and walls in cusp-decomposable manifolds. Submitted 2022. https://arxiv.org/abs/2010.03736

Teaching Experience

2024- Master's Thesis Advisor Johanna Atenea Carreón Baltazar — CCM UNAM
Thesis title "Analysis of the Distribution of Duplicated Genes in Cyanobacteria"

2024- Master's Thesis Co-advisor Luis Raul Figueroa Martinez — CCM UNAM
Thesis title "Topological Data Analysis for Automated Classification of Multicellular Patterns"

2024 Graduate course "Advanced Statistics Course: Machine Learning and Deep Geometric Learning" — CCM UNAM
https://github.com/CursosPosgradoCCM/ML-GDL

2024 Graduate course "Scientific Computing: Scientific computing for data analysis" — CCM UNAM
https://haydeeperuyero.github.io/Computo_Cientifico/

2023 Graduate course "Topology advanced course: Applications of Topological Data Analysis" — CCM UNAM
https://github.com/HaydeePeruyero/CursoA_Aplicaciones_TDA

2023 Graduate course "Statistics seminar: Data Analysis with R" — CCM UNAM
https://haydeeperuyero.github.io/Seminario_Estadistica/

Commites

2025 Master's thesis reviewer of Jaime Salvador López Viveros — CCM UNAM
Thesis title: "Identificación de recombinación en virus utilizando análisis topológicos de datos"

2025 Master's thesis reviewer of Mariel Guadalupe Gutierrez Chaveste — CCM UNAM
Thesis title: "Presentación de ADN mediante el juego de caos clasificación de imágenes y desarrollo de App de divulgación"

2024-2025 Statistics General Knowledge Basic Exam — CCM UNAM
Semester 2025-2, Semester 2025-1

2023-2024 Algebraic Topology General Knowledge Basic Exam — CCM UNAM
Semester 2023-2, 2024-1, 2024-2

Selected talks and posters

2025 CMO Workshop: Mixtures of probability and Geometry — Oaxaca, Mexico
Detecting Evolutionary Patterns and Cellular Organization through Persistent Homology

2025 La Matemática Atenagorica — Michoacan, Mexico
Diversidad microbiana en mieles de abejas nativas sin aguijón de la Península de Yucatán

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2024 — 1st Asia&Pacific Bioinformatics Joint Conference — Okinawa, Japan
Poster "Microbial Diversity and Characterization of Stingless Bee Honey from the Yucatán Peninsula"

2024 — Escuela de Matemáticas de América Latina y el Caribe Cuernavaca 2024 — Morelos, Mexico
Talk "Analizando datos con topología y geometría"

2024 — 2024 Joint Mathematics Meetings — San Francisco, USA
Talk "Asymptotic dimension and geometric decompositions in dimension 4"

2023 — XXIV Escuela de Otoño (EOBM) y el XVII Encuentro Nacional (ENBM), of Bio-mathematics — Veracruz, Mexico
Minicourse "Aplicaciones de análisis topológico de datos a Pangenómica"

2023 — Iberoamerican and Pan Pacific International Conference on Topology and its Applications — Puebla, Mexico
"Invariants under quasi-isometries"

2023 — 31st Conference on Intelligent Systems for Molecular Biology and 22th European Conference On Computational Biology as part of the Annual International Conference on Critical Assessment of Massive Data Analysis — Lyon, France
Poster "Topology, regression and metapangenome enrich conventional microbiome analyses"

2023 — 2023 Joint Mathematics Meetings — Boston, USA
Talk "Collapsing and group growth as an obstruction to Einstein metrics on some smooth 4-manifolds"

2022 — 8th Mexican Workshop on Applied Geometry and Topology — Virtual
Poster "Topological and Geometric explorations of Autism fMRI connectomes"

Recent Awards

2023 — 1 Price. Critical Assessment of Massive Data Analysis Microbiome Challenge. CAMDA 2023

2022 — Mentee in the Mentoring in Action - Women and Girls in STEM Online programme

2022 — 2nd place at the ISCB-La SOLBIO BIONETMX WIKIPEDIA COMPETITON, Article: "Matrices de distancias"

Science communication

2025 — Tianguis de las Ciencias de la Universidad Michoacana de San Nicolas de Hidalgo — Michoacan, Mexico
Workshop "Papiroflexia Modular"

2024 - 2025 — Feria de las Matemáticas — Michoacan, Mexico
Workshop "Papiroflexia Modular"

2024 — Futuras STEM Michoacán — Michoacan, Mexico
Workshop "Matemagia"

2024 — Feria Infantil y Juvenil de Ciencias y Artes — Michoacan, Mexico
Workshop "Estereogramas"

2024 — Feria de las Matemáticas — Michoacan, Mexico
Talk "¿Qué hacemos cuentas? Son puros cuentos"

2023 — 11 Fiesta de las Ciencias y Humanidades — Michoacan, Mexico
Workshop "Lengua de Señas Mexicana (LSM) y Matemáticas"

2023 — 11 Fiesta de las Ciencias y Humanidades — Michoacan, Mexico
Workshop "Biomatemáticas"

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Synergistic Activities

2024 - 2025 Encuentro de Mujeres en Topología, Geometrpia y sus Aplicaciones — Michoacan, Mexico
1st and 2nd, Co-organizer, Webpage: https://matmor.unam.mx/encuentro-mujeres-topogeo-2025/

2023 - 2025 Escuela de Teoría Geométrica de Grupos — Michoacan, Mexico
II, III, IV, Co-organizer, webpage https://matmor.unam.mx/etgg-2025/

2023 - 2025 Hackaton Reto Critical Assessment of Massive Data — Michoacan, Mexico
Co-organizer 2023, 2024 and 2025

2022- Long-distance Seminar on Geometric Group Theory in Mexico — Virtual
Co-organizer

2022- R Ladies Morelia — Hybrid, Michoacan, Mexico
Co-organizer, webpage https://www.meetup.com/es-ES/rladies-morelia/?eventOrigin=home_groups_you_organize